REDISCOVERING A FORMULA FOR π

ANSHVEER BINDRA

Abstract. This paper presents a complete and fully elementary derivation of a closed form for the one–parameter family

$$S(a) = \sum_{n=0}^{\infty} \frac{a^n}{\binom{2n+1}{n+1}}.$$

Although this series and its hypergeometric representation are already known in the literature ([2]), we give a new and self-contained proof using only high–school level methods. The argument relies solely on factorial identities, binomial coefficient manipulations, the ratio test for convergence, and the Beta–integral identity

$$\int_0^1 t^n (1-t)^n \, dt = \frac{(n!)^2}{(2n+1)!}.$$

The project originated from a computational investigation that suggested the unexpected evaluation

$$\sum_{n=0}^{\infty} \frac{2^n}{\binom{2n+1}{n+1}} = 1 + \frac{\pi}{2}.$$

We establish this identity and derive the full closed form for all $|a| < 4$ by purely elementary means. For reference, we also record that the same series admits the compact hypergeometric expression

$$S(a) = {}_2F_1\left(2, 1; \tfrac{3}{2}; \tfrac{a}{4}\right),$$

but this representation is not used in our proofs.

1. Introduction and Main Results

This project originated in a course on experimental mathematics in which students were asked to compute numerical approximations of π by hand from the series

$$\sum_{n=0}^{\infty} \frac{2^n (n!)^2}{(2n+1)!},$$

given only through its first few factored terms. During these lengthy hand computations it became natural to reorganize the summands in search of a more efficient representation. An intermediate misstep suggested that the expression might admit a compact binomial–coefficient form, which led to the investigation of the related series

$$\sum_{n=0}^{\infty} \frac{2^n}{\binom{2n+1}{n+1}}.$$

Numerical experiments then indicated that the sum appears to converge not to $\pi/2$ but to $1+\pi/2$, motivating a systematic attempt to prove the identity by purely elementary methods.

A natural extension is obtained by replacing the constant 2 with a real parameter a and considering the family

$$S(a) = \sum_{n=0}^{\infty} \frac{a^n}{\binom{2n+1}{n+1}}.$$

This generalization clarifies the observed computational patterns and leads to the central questions studied here: the domain of convergence of $S(a)$, the existence of a closed form, and the direct derivation of special values such as $S(2) = 1 + \pi/2$.

The closed form of this series is already known through its representation as a Gauss hypergeometric function, which places it within the classical theory of hypergeometric series. However, that approach typically requires familiarity with special–function machinery and transformation formulas, creating a relatively high

technical barrier to entry. The purpose of this paper is to provide a new, self-contained derivation that avoids hypergeometric theory entirely and instead uses only high–school level tools. In particular, the arguments rely on algebraic manipulations of factorials and binomial coefficients, the ratio test, and the elementary integral identity

$$\int_0^1 t^n(1-t)^n\,dt = \frac{(n!)^2}{(2n+1)!},$$

together with routine substitutions. This substantially lowers the accessibility threshold while still yielding the full closed form.

We now state the main results.

Lemma 1 (Absolute Convergence)**.** *Define*

$$A_n(a) = \frac{a^n}{\binom{2n+1}{n+1}}.$$

Then the series $\sum_{n=0}^{\infty} A_n(a)$ *converges absolutely if and only if* $|a| < 4$.

Theorem 1 (Closed–Form Evaluation)**.** *For every real* a *with* $|a| < 4$,

$$S(a) = \frac{2}{4-a} + \frac{8}{(4-a)^{3/2}\sqrt{a}}\,\arctan\!\left(\sqrt{\tfrac{a}{4-a}}\right).$$

Immediate consequences include the special values

$$S(1) = \frac{2}{3} + \frac{4\pi}{9\sqrt{3}}, \qquad S(2) = 1 + \frac{\pi}{2}, \qquad S(3) = 2 + \frac{8\pi}{3\sqrt{3}},$$

and

$$S(2-\sqrt{2}) = 2 - \sqrt{2} + \frac{\pi}{2}(\sqrt{2}-1).$$

Theorem 2 (Asymptotic Rate)**.** *If* $|a| < 4$, *then*

$$A_n(a) = \frac{\sqrt{\pi\,n}}{2}\left(\frac{a}{4}\right)^n + O\!\left(n^{-1/2}(a/4)^n\right), \qquad \lim_{n\to\infty}\frac{A_{n+1}(a)}{A_n(a)} = \frac{a}{4},$$

and $S(a)$ *converges exponentially with rate* $|a|/4$.

Corollary 2.1. *If* $0 \le |a_1| < |a_2| < 4$, *then*

$$\lim_{n\to\infty}\frac{A_{n+1}(a_1)}{A_n(a_1)} < \lim_{n\to\infty}\frac{A_{n+1}(a_2)}{A_n(a_2)},$$

so $S(a_1)$ *converges strictly faster than* $S(a_2)$.

Proof. By Theorem 2, $\lim A_{n+1}(a)/A_n(a) = |a|/4$. If $|a_1| < |a_2|$, then $|a_1|/4 < |a_2|/4$, which gives the claimed ordering of convergence rates. □

For reference, we also record the known hypergeometric representation.

Corollary 2.2 (Hypergeometric Reformulation)**.** *For all real* a,

$$S(a) = {}_2F_1\!\left(2, 1; \tfrac{3}{2}; \tfrac{a}{4}\right).$$

Proof. Using $\binom{2n+1}{n+1} = \frac{(2n+1)!}{(n+1)!\,n!}$ together with the rising factorial identities $(2)_n = 2\cdot 3\cdots(n+1)$ and $\left(\frac{3}{2}\right)_n = \frac{(2n+1)!!}{3!!}$, one verifies directly that

$$\frac{a^n}{\binom{2n+1}{n+1}} = \frac{(2)_n}{\left(\frac{3}{2}\right)_n}\left(\frac{a}{4}\right)^n,$$

which matches the defining series of ${}_2F_1$. This representation is cited only for context and is not used in the proofs. □

The remainder of the paper consists entirely of elementary derivations of the stated results.

2. Main Result and Proof

We begin by proving a key lemma.

Lemma 2. *For each integer $n \geq 0$,*

$$I_n \;\equiv\; \int_0^1 t^n (1-t)^n \, dt \;=\; \frac{(n!)^2}{(2n+1)!}.$$

Proof. Write

$$I_n = \int_0^1 t^n (1-t)^n \, dt.$$

We apply integration by parts, choosing

$$u = (1-t)^n, \quad dv = t^n \, dt,$$

so that

$$du = -n\,(1-t)^{n-1} \, dt, \qquad v = \frac{t^{n+1}}{n+1}.$$

Thus

$$I_n = (1-t)^n \, \frac{t^{n+1}}{n+1}\bigg|_0^1 - \int_0^1 \frac{t^{n+1}}{n+1} \, \left(-n\,(1-t)^{n-1}\right) \, dt.$$

The boundary term vanishes because at $t = 0$ we have $t^{n+1} = 0$ and at $t = 1$ we have $(1-t)^n = 0$. Hence

$$I_n = \frac{n}{n+1} \int_0^1 t^{n+1} (1-t)^{n-1} \, dt.$$

Repeating this argument n times, each integration by parts increases the exponent of t by one and decreases the exponent of $(1-t)$ by one, and introduces a factor

$$\frac{n-k+1}{n+k} \quad (k = 1, 2, \ldots, n).$$

After n iterations the integral reduces to

$$\int_0^1 t^{2n} \, dt = \frac{1}{2n+1},$$

while the product of the prefactors is

$$\prod_{k=1}^{n} \frac{n-k+1}{n+k} = \frac{n\,(n-1)\cdots 1}{(n+1)(n+2)\cdots(2n)} = \frac{n!}{\dfrac{(2n)!}{n!}} = \frac{(n!)^2}{(2n)!}.$$

Therefore

$$I_n = \frac{(n!)^2}{(2n)!} \, \frac{1}{2n+1} = \frac{(n!)^2}{(2n+1)!},$$

as claimed. □

We now prove the main theorem, Theorem 1

Proof. Assume $|a| < 4$; by Lemma 1 the series defining $S(a) = \sum_{n=0}^{\infty} a^n / \binom{2n+1}{n+1}$ converges absolutely on this entire open interval. We now derive a closed-form expression.

Since

$$\binom{2n+1}{n+1} = \frac{(2n+1)!}{(n+1)!\,n!},$$

we have

$$\frac{1}{\binom{2n+1}{n+1}} = \frac{(n+1)!\,n!}{(2n+1)!} = (n+1)\,\frac{n!\,n!}{(2n+1)!} = (n+1)\int_0^1 t^n (1-t)^n \, dt,$$

where the last equality follows from Lemma 2.

Therefore

$$S(a) = \sum_{n=0}^{\infty} \frac{a^n}{\binom{2n+1}{n+1}} = \sum_{n=0}^{\infty} a^n \,(n+1) \int_0^1 t^n (1-t)^n \, dt.$$

We wish to interchange summation and integration. Define

$$f_n(t) = (n+1)\,[a\,t(1-t)]^n\,, \quad t \in [0,1].$$

Then

$$|f_n(t)| = (n+1)\,|a\,t(1-t)|^n \le (n+1)\left(\frac{|a|}{4}\right)^n,$$

since $t(1-t) \le \frac{1}{4}$ on $[0,1]$. Because $|a|/4 < 1$, the series

$$\sum_{n=0}^{\infty}(n+1)\left(\tfrac{|a|}{4}\right)^n \;=\; \frac{1}{(1-|a|/4)^2}$$

converges. Since $\sum(n+1)x^n$ is a standard power series with a radius of convergence of 1, it converges uniformly on any closed interval strictly inside $(-1,1)$. Because the values of $x = a\,t(1-t)$ are bounded by $\pm|a|/4$ for all $t \in [0,1]$, they lie strictly within this interval of convergence. Thus, $\sum f_n(t)$ converges uniformly on $[0,1]$.

Uniform convergence permits term-by-term integration, yielding

$$S(a) = \int_0^1 \sum_{n=0}^{\infty}(n+1)\,[a\,t(1-t)]^n\;dt = \int_0^1 \frac{dt}{[\,1-a\,t(1-t)]^2},$$

where we have used the standard power-series identity $\sum_{n=0}^{\infty}(n+1)x^n = (1-x)^{-2}$ valid for $|x| < 1$, here with $x = a\,t(1-t)$.

We now evaluate this integral. Set

$$I(a) = \int_0^1 \frac{dt}{[1-a\,t+a\,t^2]^2}.$$

First we complete the square in the quadratic polynomial:

$$1 - a\,t + a\,t^2 = a\left(t-\tfrac{1}{2}\right)^2 + \left(1-\tfrac{a}{4}\right).$$

Introduce constants

$$c = 1 - \frac{a}{4}, \qquad d = \frac{a}{4},$$

so that

$$1 - a\,t + a\,t^2 = d\,(2t-1)^2 + c.$$

Make the substitution $u = 2t-1$; then $dt = \frac{1}{2}du$ and as t runs from 0 to 1, u runs from -1 to 1. In these variables,

$$I(a) = \int_{t=0}^1 \frac{dt}{[\,d\,(2t-1)^2 + c\,]^2} = \tfrac{1}{2}\int_{u=-1}^1 \frac{du}{(c+d\,u^2)^2} = \int_0^1 \frac{du}{(c+d\,u^2)^2},$$

where the last equality follows because the integrand is even in u.

Next, factor c from the denominator:

$$c + d\,u^2 = c\left(1 + \frac{d}{c}u^2\right) = c\left(1+\alpha^2u^2\right), \quad \alpha^2 = \frac{d}{c} = \frac{a}{4-a}.$$

Hence

$$I(a) = \frac{1}{c^2}\int_0^1 \frac{du}{(1+\alpha^2u^2)^2}.$$

Perform the change of variable $w = \alpha\,u$, so $dw = \alpha\,du$ and $u \in [0,1] \iff w \in [0,\alpha]$. Then

$$\int_0^1 \frac{du}{(1+\alpha^2u^2)^2} = \frac{1}{\alpha}\int_0^\alpha \frac{dw}{(1+w^2)^2}.$$

We recall the standard antiderivative:

$$\int \frac{1}{(1+w^2)^2}dw = \tfrac{w}{2(1+w^2)} + \tfrac{1}{2}\arctan w.$$

Therefore

$$\int_0^\alpha \frac{dw}{(1+w^2)^2} = \left[\frac{w}{2(1+w^2)} + \frac{1}{2}\arctan w\right]_{w=0}^{w=\alpha} = \frac{\alpha}{2(1+\alpha^2)} + \frac{1}{2}\arctan(\alpha).$$

Putting these pieces together,

$$I(a) = \frac{1}{c^2}\,\frac{1}{\alpha}\left(\frac{\alpha}{2(1+\alpha^2)} + \frac{1}{2}\arctan(\alpha)\right) = \frac{1}{c^2}\left(\frac{1}{2(1+\alpha^2)} + \frac{1}{2\alpha}\arctan(\alpha)\right).$$

Finally, substitute back $c = (4-a)/4$ and $\alpha^2 = a/(4-a)$. Then

$$\frac{1}{2(1+\alpha^2)} = \frac{4-a}{8}, \quad \frac{1}{2\alpha}\arctan(\alpha) = \frac{\sqrt{4-a}}{2\sqrt{a}}\ \arctan\!\left(\sqrt{\tfrac{a}{4-a}}\right), \quad c^2 = \frac{(4-a)^2}{16}.$$

Hence

$$I(a) = \frac{16}{(4-a)^2}\left[\frac{4-a}{8} + \frac{\sqrt{4-a}}{2\sqrt{a}}\arctan\!\left(\sqrt{\tfrac{a}{4-a}}\right)\right] = \frac{2}{4-a} + \frac{8}{(4-a)^{3/2}\sqrt{a}}\ \arctan\!\left(\sqrt{\tfrac{a}{4-a}}\right).$$

Since $S(a) = I(a)$, this completes the proof of the closed-form formula. □

3. Rate of Convergence

We now provide a proof for Theorem 2.

Proof. We begin by obtaining an asymptotic approximation for the central binomial coefficient $\binom{2n}{n}$. By Stirling's formula [7, 3], valid for large n,

$$n! = n^n e^{-n}\sqrt{2\pi n}\ (1 + O(1/n)).$$

Hence

$$(2n)! = (2n)^{2n}e^{-2n}\sqrt{4\pi n}\ (1 + O(1/n)) = 2^{2n}n^{2n}e^{-2n}\sqrt{4\pi n}\ (1 + O(1/n)).$$

On the other hand,

$$(n!)^2 = n^{2n}e^{-2n}(2\pi n)\ (1 + O(1/n)).$$

Dividing these expressions gives

$$\binom{2n}{n} = \frac{(2n)!}{(n!)^2} = \frac{2^{2n}n^{2n}e^{-2n}\sqrt{4\pi n}\,(1 + O(1/n))}{n^{2n}e^{-2n}(2\pi n)\,(1 + O(1/n))} = \frac{2^{2n}}{\sqrt{\pi n}}\,(1 + O(1/n)).$$

Next, observe that

$$\binom{2n+1}{n+1} = \frac{(2n+1)!}{(n+1)!\,n!} = \frac{2n+1}{n+1}\binom{2n}{n}.$$

Since

$$\frac{2n+1}{n+1} = 2\,\frac{n+\frac{1}{2}}{n+1} = 2\left(1 - \tfrac{1}{2(n+1)}\right) = 2\ (1 + O(1/n)),$$

we conclude

$$\binom{2n+1}{n+1} = 2\binom{2n}{n}\ (1 + O(1/n)) = \frac{2^{2n+1}}{\sqrt{\pi n}}\,(1 + O(1/n)).$$

Therefore the general term of our series is

$$A_n(a) = \frac{a^n}{\binom{2n+1}{n+1}} = a^n\,\frac{\sqrt{\pi n}}{2^{2n+1}}\ (1 + O(1/n))^{-1}.$$

Because $(1 + O(1/n))^{-1} = 1 + O(1/n)$, we obtain the precise asymptotic expansion

$$A_n(a) = \frac{\sqrt{\pi n}}{2}\left(\frac{a}{4}\right)^n (1 + O(1/n)),$$

or equivalently

$$A_n(a) = \frac{\sqrt{\pi n}}{2}\left(\frac{a}{4}\right)^n + O\!\left(n^{-\frac{1}{2}}(a/4)^n\right).$$

This shows that the dominant behavior is geometric, with factor $(a/4)^n$, modulated by a slowly varying $\sqrt{n}$ term.

To find the limit of successive-term ratios, divide

$$A_{n+1}(a) = \frac{\sqrt{\pi(n+1)}}{2}\left(\tfrac{a}{4}\right)^{n+1}(1 + O(1/(n+1)))$$

by

$$A_n(a) = \frac{\sqrt{\pi n}}{2} \left(\tfrac{a}{4}\right)^n (1 + O(1/n)).$$

Cancelling common factors yields

$$\frac{A_{n+1}(a)}{A_n(a)} = \frac{a}{4} \sqrt{\frac{n+1}{n}} \, \frac{1 + O(1/(n+1))}{1 + O(1/n)} = \frac{a}{4} (1 + O(1/n)),$$

and therefore

$$\lim_{n\to\infty} \frac{A_{n+1}(a)}{A_n(a)} = \frac{a}{4}.$$

Since $|a/4| < 1$, the Ratio Test confirms that the series converges absolutely and that its terms decay exponentially at rate $|a/4|$.

Finally, we estimate the tail (remainder) after N terms. Define

$$R_N(a) = \sum_{n=N+1}^{\infty} A_n(a).$$

Choose $\varepsilon > 0$ small enough that $|a/4| + \varepsilon < 1$. Then for sufficiently large n, say $n \geq N_0$, the ratio satisfies

$$\left| \frac{A_{n+1}(a)}{A_n(a)} \right| \leq |a/4| + \varepsilon.$$

Hence for all $k \geq 1$ and $N \geq N_0$,

$$|A_{N+k}(a)| \leq (|a/4| + \varepsilon)^{k-1} \, |A_{N+1}(a)|.$$

Thus

$$R_N(a) \leq |A_{N+1}(a)| \sum_{k=0}^{\infty} (|a/4| + \varepsilon)^k = \frac{|A_{N+1}(a)|}{1 - (|a/4| + \varepsilon)}.$$

Since

$$A_{N+1}(a) \sim \frac{\sqrt{\pi(N+1)}}{2} \left(\frac{|a|}{4}\right)^{N+1},$$

we deduce

$$R_N(a) = O\Big(\sqrt{N} \, (|a|/4)^N\Big).$$

Finally, by the same ratio argument,

$$\lim_{N\to\infty} \frac{R_{N+1}(a)}{R_N(a)} = \frac{|a|}{4}.$$

This establishes the claimed rate of convergence and completes the proof. □

To illustrate the geometric rate of convergence established in Theorem 2, we compute the partial sums' rearrangement to yield π. We define

$$S_n(a) \;=\; \sum_{k=0}^{n} \frac{a^k}{\binom{2k+1}{k+1}}, \qquad S(a) \;=\; C_a + k_a \, \pi,$$

with

$$\begin{aligned} C_1 &= \tfrac{2}{3}, & k_1 &= \frac{4}{9\sqrt{3}}, \\ C_2 &= 1, & k_2 &= \tfrac{1}{2}, \\ C_3 &= 2, & k_3 &= \frac{8}{3\sqrt{3}}, \\ C_{2-\sqrt{2}} &= 2 - \sqrt{2}, & k_{2-\sqrt{2}} &= \frac{\sqrt{2}-1}{2}. \end{aligned}$$

Then

$$\pi_n(a) \;=\; \frac{S_n(a) - C_a}{k_a} \;\longrightarrow\; \pi.$$

We compute $\pi_n(a)$ for $n = 5, 6, 7, 8, 9, 50$:

n	$\pi_n(1)$	$\pi_n(2)$	$\pi_n(3)$	$\pi_n(2-\sqrt{2})$
5	3.1384985	2.982972	1.844802	3.1414579
6	3.1407696	3.057564	2.120734	3.1415716
7	3.1413752	3.097347	2.341480	3.1415893
8	3.1415355	3.118408	2.516778	3.1415921
9	3.1415777	3.129493	2.655171	3.1415926
50	3.1415926	3.141592	3.141585	3.1415926

TABLE 1. Estimates $\pi_n(a) = (S_n(a) - C_a)/k_a$ for $n = 5, \ldots, 9, 50$.

As Theorem 2 predicts, the error $|\pi - \pi_n(a)|$ decays geometrically at rate $\rho = |a|/4$. For $a = 2 - \sqrt{2}$, $\rho \approx 0.1464$ yields seven-figure accuracy by $n = 8$. In contrast, $a = 1, 2, 3$ have $\rho = 1/4$, $2/4$, $3/4$ respectively, so they require more terms to attain the same precision.

For context, the classical Madhava–Leibniz (also known as Gregory–Leibniz) series for π is

$$\frac{\pi}{4} = \sum_{n=0}^{\infty} \frac{(-1)^n}{2n+1}.$$

Its nth term in absolute value is $1/(2n+1)$, so the ratio of successive magnitudes satisfies

$$\frac{1/(2(n+1)+1)}{1/(2n+1)} = \frac{2n+1}{2n+3} \longrightarrow 1 \quad \text{as } n \to \infty.$$

Consequently, the Leibniz series converges only *subpolynomially* (indeed, very slowly), requiring on the order of 10^k terms to achieve k correct decimal places of π [6, 5, 1]. In stark contrast, our binomial-series expansion has general term $A_n = a^n / \binom{2n+1}{n+1}$, whose consecutive-term ratio tends to $|a|/4$. This geometric decay means each term roughly halves the remaining error, so after N terms the error R_N is on the order of $(|a|/4)^N$ (up to a mild $\sqrt{N}$ factor). Thus our series achieves exponentially fast convergence, vastly outperforming the Madhava–Leibniz expansion.

4. Conclusion

The purpose of this paper was to evaluate the family of series

$$S(a) = \sum_{n=0}^{\infty} \frac{a^n}{\binom{2n+1}{n+1}}, \qquad |a| < 4,$$

by relying only on elementary arguments. The guiding philosophy was that the closed form of $S(a)$ can be reached without appealing to complex analysis, advanced special functions, or anything beyond the standard toolkit of high–school mathematics.

The central ideas were simple: factorial identities, straightforward manipulations of binomial coefficients, the ratio test for convergence, and the Beta–function integral

$$\int_0^1 t^n (1-t)^n \, dt = \frac{(n!)^2}{(2n+1)!},$$

which itself follows from two elementary integrations by parts. These tools were sufficient to rewrite each term of $S(a)$ as an integral, justify interchanging summation and integration, and reduce the problem to the evaluation of a single explicit integral. The resulting computation produced the closed form

$$S(a) = \frac{2}{4-a} + \frac{8}{(4-a)^{3/2}\sqrt{a}} \arctan\left(\sqrt{\tfrac{a}{4-a}}\right),$$

from which the special identity

$$S(2) = 1 + \frac{\pi}{2}$$

follows immediately. We also quantified the exponential rate of convergence of $S(a)$ and its dependence on $|a|$, again using only elementary asymptotic estimates.

Although the derivation presented here is wholly elementary, it is noteworthy that the family $S(a)$ admits a concise formulation in classical notation:

$$S(a) = {}_2F_1\big(2, 1; \tfrac{3}{2}; \tfrac{a}{4}\big)\,.$$

This hypergeometric representation plays no role in the proofs, but it situates the result within a long tradition of identities evaluated by Euler, Gauss, and later authors. In this sense, the elementary arguments of the paper provide an alternative route to a classical evaluation, showing that ideas usually encountered in advanced courses can emerge naturally from experimentation and careful manipulation.

The broader perspective is educational. A simple computational exercise led to a conjectured identity for $S(2)$, which in turn motivated the introduction of a parameter and the search for a general formula. The fact that such a formula could be reached using only high–school methods emphasizes that significant mathematical structure is often accessible far earlier than one might expect. The derivation of $S(a)$ illustrates how pattern recognition, systematic experimentation, and elementary reasoning can converge to reproduce results that traditionally appear in much more sophisticated settings.

Finally, this exploration shows that rediscovery is a natural part of learning mathematics. Even when an identity is classical, approaching it from an elementary standpoint offers insight into why it holds, how it fits together, and how deeper structures can surface from the simplest of calculations. In this way, $S(a)$ serves not only as an instructive example of an explicitly summable series, but also as a reminder of the power and elegance of elementary methods.

5. Acknowledgments

I would like to express my sincere gratitude to Sagar Shrivastava for his invaluable guidance throughout the preparation of this paper. His insights, detailed feedback, and meticulous proofreading have significantly improved the clarity and rigor of my work. I am deeply appreciative of his encouragement and support at every stage of this project.

I also wish to thank Anubhav Bhattacharjee for his collaborative spirit and for helping me discover the initial form of the series. Our numerous discussions and his thoughtful proofreading have been instrumental in refining the arguments presented here. Without his contributions, this work would not have been possible.

References

[1] Mark B. Villarino and. "The Error in an Alternating Series". In: *The American Mathematical Monthly* 125.4 (2018), pp. 360–364. DOI: 10.1080/00029890.2017.1416875. eprint: https://doi.org/10.1080/00029890.2017.1416875. URL: https://doi.org/10.1080/00029890.2017.1416875.

[2] Wilfrid Norman Bailey. *Generalized Hypergeometric Series*. Vol. 32. Cambridge Tracts in Mathematics and Mathematical Physics. Cambridge, England: Cambridge University Press, 1935, p. 108. ISBN: 0598959017.

[3] Robert Sachs. *Calculus Favorite: Stirling's Approximation, Approximately*. 2011. URL: https://math.gmu.edu/~rsachs/talks/maastirlingbeamer.pdf.

[4] Wikipedia contributors. *Beta function — Wikipedia, The Free Encyclopedia*. 2025. URL: https://en.wikipedia.org/w/index.php?title=Beta_function&oldid=1285867538.

[5] Wikipedia contributors. *Leibniz formula for π — Wikipedia, The Free Encyclopedia*. 2025. URL: https://en.wikipedia.org/w/index.php?title=Leibniz_formula_for_%CF%80&oldid=1285636583.

[6] Wikipedia contributors. *Pi — Wikipedia, The Free Encyclopedia*. 2025. URL: https://en.wikipedia.org/w/index.php?title=Pi&oldid=1292836233.

[7] Wikipedia contributors. *Stirling's approximation — Wikipedia, The Free Encyclopedia*. 2025. URL: https://en.wikipedia.org/w/index.php?title=Stirling%27s_approximation&oldid=1286472739.